

COCA-COLA AFRICA

South Africa — Media Sr. Director

Media Analytics Dashboard

Detailed Scope & Workflow Document | Prototype Phase

1. Executive Summary

The Media Analytics Dashboard is a drill-down intelligence tool designed for Coca-Cola Africa's South Africa business unit. It enables brand, sales, and trade marketing teams to monitor TCCC brand performance (value and volume share) and rapidly identify the root cause of any share loss — from overall brand level down to individual region and causal factor — in as few as five clicks.

In addition to the manual drill-down journey, the dashboard features a natural-language AI prompt interface. A user can simply type a question such as "How is Sparletta performing in South Africa?" and receive an instant, automatically generated summary of all five analytical layers — eliminating the need to navigate manually.

2. Project Objectives

- Provide real-time visibility into TCCC brand value share and volume share across South Africa.
- Enable rapid root-cause analysis of share movements through an intuitive click-through hierarchy.
- Reduce analytical turnaround time — from days to seconds — for brand and trade teams.
- Deliver an AI-powered natural language interface that automates the full analytical journey.
- Build a working prototype with dummy data to validate feasibility before full deployment.

3. Project Scope

3.1 In Scope

- Prototype dashboard with 5-level drill-down hierarchy (Brand → Channel → Customer → Region → Root Cause).
- Value share and volume share visualisations for all TCCC brands.
- Channel breakdown: TEG, PFM, L&T, HORECA.
- Customer/retailer breakdown: Shoprite, Woolworths, Pick n Pay, and others.
- Regional breakdown: Gauteng, KZN, Western Cape, and other South African regions.
- Root cause diagnostics layer: distribution, pricing (own and competitor), promotions, out-of-stock.
- AI natural language prompt interface for automated end-to-end summary generation.
- Dummy / synthetic data for the prototype phase.

3.2 Out of Scope (Prototype Phase)

- Live data integration with Coca-Cola Africa's data systems.
- User authentication and access control (role-based permissions).

- Export to PDF / PowerPoint (can be added post-prototype).
- Mobile-responsive design (desktop-first for prototype).
- Historical trending beyond what dummy data supports.

4. Data Architecture

The dashboard operates on a hierarchical data model. Each level of the drill-down corresponds to a different data dimension:

Level	Data Dimension & Description
Level 1 — Brand	Monthly value share & volume share by TCCC brand (e.g., Sparletta, Coca-Cola, Fanta).
Level 2 — Channel	Share broken down by trade channel: TEG (Traditional Grocery), PFM (Petrol/Forecourt), L&T (Liquor & Tobacco), HORECA (Hotels/Restaurants/Cafes).
Level 3 — Customer	Share by key retail account within the selected channel (Shoprite, Woolworths, Pick n Pay, Spar, etc.).
Level 4 — Region	Share by South African province/region (Gauteng, KZN, Western Cape, Eastern Cape, etc.).
Level 5 — Root Cause	Causal factor flags: distribution gap, price increase (own), price decrease (competitor), promo absence, out-of-stock event.

5. Detailed Drill-Down Workflow

The dashboard follows a linear drill-down navigation model. Each step narrows the analysis from a broader view to a more specific one. The user can exit and re-enter the hierarchy at any level.

STEP 1	Brand-Level Share Overview — The Starting Point
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What the user sees:

- A bar or line chart showing Value Share (%) and Volume Share (%) for all major TCCC brands, plotted by month.
- Brands displayed: Sparletta, Coca-Cola, Fanta, Sprite, Powerade, Schweppes, etc.
- Time axis: last 12 months, with current month highlighted.
- Ability to toggle between Value Share and Volume Share views.

User Action:

User notices Sparletta's share has dropped in March. They click on the Sparletta bar for March.

What is triggered:

Dashboard filters all subsequent data to Sparletta / March and navigates to Step 2.

**STEP
2****Channel Breakdown — Where is the Drop Coming From?****What the user sees:**

- A breakdown of Sparletta's March share performance split by the four trade channels.
 - TEG (Traditional / General Trade)
 - PFM (Petrol, Forecourt & Mobility)
 - L&T (Liquor & Tobacco)
 - HORECA (Hotels, Restaurants, Cafes)
- Visual indicators (red/green arrows) showing which channels are over- or under-performing vs prior month.

User Action:

TEG is showing a significant red indicator — the user clicks on TEG.

What is triggered:

Dashboard filters to Sparletta / March / TEG channel and navigates to Step 3.

**STEP
3****Customer / Retailer Breakdown — Which Account is the Problem?****What the user sees:**

- A ranked list or bar chart of Sparletta's share within TEG, broken down by key retail accounts.
 - Shoprite / Checkers
 - Pick n Pay
 - Woolworths
 - Spar
 - Boxer / Others
- Share for current period (March) vs prior period (February), with delta displayed.

User Action:

Shoprite shows a sharp decline. The user clicks on Shoprite.

What is triggered:

Dashboard filters to Sparletta / March / TEG / Shoprite and navigates to Step 4.

**STEP
4****Regional Breakdown — In Which Geography is the Drop Concentrated?**

What the user sees:

- An interactive South Africa map OR a ranked table showing Sparletta's share at Shoprite by province/region.
 - Gauteng
 - KwaZulu-Natal (KZN)
 - Western Cape (WC)
 - Eastern Cape (EC)
 - Limpopo / Mpumalanga / Free State / North West
- Heat-map colouring: dark red = worst performing region, green = stable/growing.

User Action:

KZN is the darkest red region. The user clicks on KZN.

What is triggered:

Dashboard filters to Sparletta / March / TEG / Shoprite / KZN and navigates to Step 5.

STEP
5

Root Cause Diagnostics — Why is the Drop Happening?

What the user sees:

A diagnostics panel displaying the key causal factors that explain the share drop in KZN / Shoprite / TEG for Sparletta in March. Each factor is flagged as a contributor or non-contributor:

Root Cause Factor	Detail / Example Output
Distribution Drop	Sparletta SKU availability in KZN Shoprite stores dropped from 87% to 63% in March.
Own Price Increase	Sparletta average shelf price increased by +8.2% vs February in KZN.
Competitor Price Drop	Key competitor (e.g., Soda brand X) reduced price by -6% in KZN during the same period.
Promotional Gap	No Sparletta promotions ran in KZN Shoprite in March vs a competitor promo in the same window.
Out-of-Stock Events	OOS events recorded on 9 of 28 days in March for key Sparletta SKUs in KZN Shoprite.

Output:

The dashboard surfaces the top 1–3 root causes ranked by estimated share impact, giving the commercial team a clear, actionable picture of what needs to be fixed and where.

6. AI Natural Language Prompt Interface

This is the most powerful feature of the dashboard. Instead of navigating through all five steps manually, the user types a plain-English question into a prompt box.

6.1 How it Works

Stage	Description
User Input	User types a question, e.g.: "How is Sparletta performing in South Africa and what are the draggers and drivers?"
AI Processing	The natural language prompt is parsed and mapped to the 5-layer data model. The AI engine queries all relevant data dimensions simultaneously.
Output Generation	A structured, plain-English summary is generated covering: overall share position, channel performance, key customer performance, regional performance, and root cause diagnosis.
Visual Support	Mini charts and highlight tables accompany the text summary for quick visual scanning.

6.2 Example Output (AI Prompt Response)

Example Question typed by user:

"How is Sparletta performing in South Africa and what are the draggers and drivers of performance?"

AI-generated summary output would cover:

- Overall Share: Sparletta value share at X% in March, down Y bps vs February.
- Channel View: TEG is the primary drag (-Z bps); PFM and HORECA are stable.
- Customer View: Shoprite accounts for 70% of the TEG decline; Pick n Pay is flat.
- Regional View: KZN and Gauteng are the drag regions; Western Cape is positive.
- Root Cause: Drop driven primarily by distribution loss and a competitor price promotion in KZN.
- Recommended Action (optional): Prioritise distribution recovery in KZN Shoprite and review pricing competitiveness.

7. Proposed Technology Stack

Component	Recommended Technology
Frontend Framework	React.js — for interactive, component-based UI
Charting Library	Recharts or Chart.js — for bar charts, line charts, and share trend visuals
Map Visualisation	Leaflet.js or Mapbox — for the South Africa regional heatmap (Step 4)
AI Prompt Engine	Claude API (Anthropic) — for natural language query processing and summary generation
Dummy Data Layer	JSON / CSV flat files for the prototype; easily replaceable with live API in production
State Management	React Context or Redux — to maintain drill-down filter state across all 5 levels
Styling	Tailwind CSS — for clean, professional dashboard aesthetics
Deployment (Proto)	Vercel or Netlify — for fast prototype hosting and sharing with stakeholders

8. Dummy Data Structure

For the prototype phase, synthetic (dummy) data will be created to simulate realistic market conditions. The data will be structured in the following format:

- Brands: Sparletta, Coca-Cola Original, Fanta Orange, Sprite, Powerade (5 TCCC brands + 2 competitor brands).
- Time Period: January 2024 – March 2024 (3 months to show trends and a clear March dip for Sparletta).
- Channels: TEG, PFM, L&T, HORECA.
- Customers: Shoprite, Pick n Pay, Woolworths, Spar, Boxer.
- Regions: Gauteng, KZN, Western Cape, Eastern Cape, Limpopo.
- Root Cause Flags: Distribution %, Average Shelf Price (own & competitor), Promo flag (Y/N), OOS Days.

9. Key Assumptions & Risks

9.1 Assumptions

- All data for the prototype will be dummy/synthetic — no live data integration required at this stage.
- The prototype is desktop-first; mobile optimisation can be addressed in Phase 2.
- Stakeholders will review and provide feedback within 5 business days of demo.
- Final data structure and field names will be confirmed by Coca-Cola Africa team before build commences.

9.2 Risks & Mitigations

Risk	Mitigation
Dummy data may not reflect real market patterns accurately	Coca-Cola team to validate dummy data ranges before build.
AI prompt output quality may vary for complex questions	Define a constrained prompt template for the prototype phase.
Scope creep during prototype build	Strictly follow in-scope list; log additional requests for Phase 2.
Drill-down UX may feel unintuitive to some users	Conduct a 30-minute usability test with 2–3 end users post-build.

10. Success Criteria for Prototype

- All 5 drill-down steps function correctly and navigate in sequence without errors.
- AI prompt produces a coherent, accurate summary of all 5 analytical layers in under 10 seconds.
- Stakeholders from Coca-Cola Africa Media Sr. Director team confirm the prototype meets the original brief.
- Prototype is approved for Phase 2 (live data integration and full production build).

11. Immediate Next Steps

1. Coca-Cola Africa team to review and approve this scope document.
2. Share any additional wireframe references or design preferences.

3. Confirm the dummy data structure and approve the synthetic dataset.
4. Agree on stakeholder review cadence (weekly check-ins recommended).
5. Kick off prototype build — Week 1 commences upon scope sign-off.

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